

Project Assignment

Overview

The course includes a semester-long project completed in teams of **no more than three (3) students**. Teams must be self-formed, and each team is responsible for selecting its own project topic. All team members will receive the same project grade. The instructional team will not arbitrate internal team disputes, so students are expected to manage collaboration, workload distribution, and communication professionally.

Project Scope and Expectations: Each team will identify and study a meaningful real-world problem that can be addressed using machine learning techniques. Given the Industrial Engineering focus of the course, students are strongly encouraged to select problems related to areas such as operations, manufacturing, supply chains, logistics, healthcare systems, energy systems, transportation, human-centered systems, or other socio-technical systems of practical importance. Projects should be well-motivated, clearly justified, and worth sustained effort. Teams are encouraged to pursue pressing or impactful problems whose analysis may yield useful insights, inform decision-making, or serve as a foundation for a future capstone project, internship application, or research effort.

Data Requirements: Projects must include a data-driven experimental component. Teams are expected to identify and work with an appropriate dataset, preferably real-world data. Synthetic or simulated data may be used when real data are unavailable or inappropriate, but the motivation for doing so must be clearly explained. Students should demonstrate thoughtful data preprocessing, modeling choices, evaluation, and interpretation of results.

Analysis and Outcomes: Projects should go beyond simply applying algorithms. Teams are expected to clearly articulate the problem context, justify modeling decisions, analyze results critically, and discuss limitations and potential improvements. Emphasis should be placed on insights gained, implications for the underlying system, and directions for future work.

Project Deliverables: Each team is required to submit the following components:

- **Proposal:** A mid-semester project proposal outlining the problem, data source(s), and planned methodology
- **Final Submission:** A final written project report documenting the problem formulation, methods, experiments, results, and conclusions
- **Peer Evaluation:** A *confidential* peer evaluation assessing individual contributions within the team

What Makes a Good ISYE 4600 Project?

This is not a “try a model on a dataset” exercise. Your project should resemble a small applied research or analytics study. A strong project:

- Starts from a **real decision, risk, or system behavior** that matters.
- Uses machine learning as a **tool to answer an important question**, not as the goal itself.

- Is **data-driven, quantitative, and reproducible**.
- Includes careful evaluation, not just model fitting.
- Produces **insights that would matter to a stakeholder**.

Examples of good problem framings:

- “Can we predict emergency room overcrowding 6 hours in advance so hospitals can reallocate staff?”
- “Which suppliers are most likely to fail to deliver on time, and why?”
- “Which customers are at highest risk of churn, and what features drive that risk?”
- “Can we detect abnormal energy usage that indicates equipment failure?”

Examples of weak framings:

- “We will apply k-means to a dataset.”
- “We will try several classifiers and see which is best.”
- “We will analyze some data from Kaggle.”

Deliverable 1: Mid-Semester Project Proposal

The proposal is **not** a vague idea or a one-page sketch. It is a **commitment to a concrete, feasible, and meaningful project**. You are expected to have already found your data and designed your approach.

Required Sections

Your proposal must include the following sections:

1. Problem Statement

Describe:

- The real-world system or decision being studied
- Who cares about this problem and why
- What goes wrong today without good predictions or decisions

This should be specific and grounded in reality, not abstract.

2. Research Questions or Objectives

State clearly what you are trying to learn or predict. Examples:

- “Which patients are most likely to be readmitted within 30 days?”
- “Can we forecast demand well enough to reduce inventory waste by 20%?”

3. Dataset Description

You must identify your dataset in the proposal. Include:

- Source (URL, database, organization, API, etc.)
- Unit of analysis, key features and the target variable (if supervised)
- Size of the dataset (rows, columns; time range if relevant)

If the data are synthetic, explain why real data are not available and how the data were generated.

4. Proposed Machine Learning Methods

Describe:

- At least one simple baseline (required)
- One or two additional methods you will compare against the baseline
- Why these choices make sense for your problem and data

5. Evaluation Plan

Explain:

- Metrics you will use (e.g., accuracy, F1, RMSE/MAE, silhouette score)
- Validation plan (train/validation/test split or cross-validation)
- What success looks like (e.g., improvement over baseline; useful predictions)

6. Action Plan

Provide a short checklist with owners and dates. At minimum include:

- Data acquisition + data dictionary
- Cleaning/preprocessing plan
- Baseline implementation
- Model comparisons + tuning plan
- Final report writing plan

Deliverable 2: Final Project Report

The final report should be a clear, well-organized applied ML write-up that demonstrates you can (i) frame a real IE-relevant problem, (ii) work with data, (iii) build and evaluate models responsibly, and (iv) communicate results. It does **not** need to be novel research.

Submission Package

Submit one zipped folder containing four **required** elements:

- Report PDF
- Code (notebook(s) and/or .py files)
- Data *or* a clear link + download instructions (if data cannot be redistributed)
- README with step-by-step instructions to reproduce results

Recommended Report Structure

1. **Problem Statement and Goal.** What decision or question are you addressing? Why is it important? Who benefits?
2. **Data.** Source, unit of analysis, target label/response (if any), key features, preprocessing, and train/test split.

3. **Methods.** Baseline + comparison model(s). Brief description of methods, data processing, and any contributions.
4. **Evaluation.** Metrics, validation strategy, and why they fit the problem.
5. **Results.** Performance tables and figures followed by brief explanation; comparisons to baselines; simple error analysis (what cases fail?).
6. **Interpretation for the System.** What would you recommend a stakeholder do with these results? What are risks or assumptions?
7. **Limitations and Next Steps.** Data limits, generalization concerns, and one or two realistic improvements.

Minimum Expectations for Technical Content

- Include a baseline and at least one stronger method, with a fair comparison.
- Include at least one clear table or figure that summarizes performance.
- Use a principled evaluation strategy (do not evaluate on training data).
- Include brief discussion of what the model gets wrong (error analysis) and why it matters.

Rubric

Projects will be graded based on:

- Quality and importance of the problem
- Soundness of methodology
- Use of data and evaluation
- Depth of analysis
- Clarity of writing and figures
- Reproducibility

A technically correct but trivial project will not score highly. A thoughtful, well-justified, carefully analyzed project will.

Proposal (25 points)

- **Problem clarity and significance (7):** Specific IE-relevant problem, clear stakeholder/impact.
- **Dataset (7):** Concrete source, unit of analysis, size, target/features clearly stated.
- **Methodology (6):** Baseline + planned comparisons are appropriate and justified.
- **Evaluation + action plan (5):** Metrics/validation are reasonable; actionable task list with responsibilities/timeline.

Final Report (65 points)

- **Problem, framing, and objectives (10):** Clear goal and research questions/objectives; appropriate scope for the course.
- **Data and preprocessing (10):** Data description, cleaning steps, feature/target definition, split strategy, limitations/biases.

- **Methods and implementation (10):** Baseline + comparison models; feature engineering; tuning described clearly.
- **Evaluation quality (10):** Correct metrics, validation, and fair comparisons; results not based on training performance.
- **Results and analysis (15):** Performance summary (tables/figures), comparison to baseline, and meaningful error analysis.
- **Communication (5):** Clear writing, well-labeled figures/tables, coherent narrative.
- **Peer evaluation (5):** A *confidential* peer evaluation assessing individual contributions within the team

Reproducibility & Submission Quality (10 points)

- **Complete package (6):** Code + data (or links) + report submitted as required.
- **README and reproducibility (4):** Clear instructions that allow a TA to rerun and reproduce key results.